

LESSON**7****Split Arcade Drive***One stick drives, one stick turns***MISSION** Can you mix and normalize drive and turn while preserving steering?**Learning targets**

CHECK	BY THE END OF THIS LESSON, I CAN...
[]	Compare tank and split arcade drive.
[]	Identify drive and turn axes.
[]	Compute left/right mixes.
[]	Normalize an out-of-range result.

Words to know**VOCABULARY** split arcade | drive | turn | mix | Math.abs | Math.max | denominator | normalize | ratio**Code focus**

```
double drive = -gamepad1.left_stick_y;
double turn = gamepad1.right_stick_x;
double denominator = Math.max(Math.abs(drive) + Math.abs(turn), 1.0);
double leftPower = (drive + turn) / denominator;
double rightPower = (drive - turn) / denominator;
```

Before the video: commit to a prediction**MY PREDICTION** For drive = +0.5 and turn = +0.5, calculate the raw left and right mixes.

Confidence before the lesson: 1 2 3 4 5 (circle one)

The POWERING ON safety contract

NON-NEGOTIABLE The sequence is completed before every live-motor test. If a step is skipped, remain disabled and restart at Step 1.

STEP	ACTION	TEAM CHECK
1	SECURE THE ROBOT Use stable blocks. Keep driven wheels and moving mechanisms clear of the table and floor.	[]
2	CONFIRM DISABLED Select the correct robot and OpMode, but do not press PLAY.	[]
3	LOOK AROUND Remove hands, tools, wires, scraps, and loose objects from moving areas.	[]
4	MAKE SURE ALL ARE CLEAR Everyone steps outside the boundary; secure hair, clothing, lanyards, strings, and jewelry.	[]
5	CALL AND WAIT Call POWERING ON! loudly. Pause long enough for everyone to stop, look up, and step away.	[]
6	VISUALLY CONFIRM, THEN ENABLE Scan the full area again. Press PLAY only when everyone and everything is clear.	[]

Callout script

BEFORE PLAY Robot is secured and DISABLED. Look around. Everyone clear? POWERING ON! ... Wait ... Area is visibly clear. Enabling now.

BEFORE CONTACT STOP pressed. Driver Station shows DISABLED. Hands may approach only after the operator confirms disabled.

Student agreement

I understand that anyone may call STOP and the operator will disable immediately. I will not approach or touch the robot until DISABLED is visually confirmed.

Student signature: _____ Date: _____

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Guided Video Notes

Split Arcade Drive

DIRECTIONS At each timestamp, pause, think silently, discuss, and then record evidence from the code or whiteboard visual.

PAUSE	QUESTION	MY CODE / VISUAL EVIDENCE
0:18	How do tank and split arcade drive differ?	
0:28	Which axis supplies drive, and why is it negated?	
0:38	Which axis supplies turn?	
0:48	Why does one formula add turn while the other subtracts?	
1:08	How can mixed values exceed the motor-power range?	
1:18	What does normalization accomplish?	

MOST IMPORTANT IDEA The idea I would teach to a teammate is:

LESSON**7****Hands-On Team Activity***Split Arcade Drive***Materials and setup**

READY	ITEM
[]	BasicArcadeDriveTeleOp.java or the five key split-arcade lines displayed.
[]	Role cards: DRIVE, TURN, ADD, SUBTRACT, DENOMINATOR, LEFT POWER, RIGHT POWER, and MOTION.
[]	Value cards including 0.0, +0.2, +0.5, +0.6, and +1.0.
[]	Printed mix-and-normalize worksheet from this guide.
[]	Optional secured robot with Drive, Turn, Left Power, and Right Power telemetry.

Team roles

ROLE	JOB
DRIVE	Holds <code>-gamepad1.left_stick_y</code> .
TURN	Holds <code>gamepad1.right_stick_x</code> .
ADD	Calculates <code>drive + turn</code> .
SUBTRACT	Calculates <code>drive - turn</code> .
DENOMINATOR	Calculates <code>max(abs(drive)+abs(turn), 1.0)</code> .
LEFT POWER	Divides the left raw mix by the denominator.
RIGHT POWER	Divides the right raw mix by the denominator.
MOTION	Predicts the resulting robot movement.

Activity sequence

STEP	ACTION
1	Ask students to show with their hands how tank drive and split arcade assign joystick jobs.
2	Set <code>drive = +0.5</code> and <code>turn = 0.0</code> . Walk through raw left, raw right, denominator, and final outputs.
3	Set <code>drive = 0.0</code> and <code>turn = +0.5</code> . Require both formulas to be evaluated.
4	Set <code>drive = +1.0</code> and <code>turn = +1.0</code> . Do not skip the raw-mix step.
5	Students solve <code>drive = +0.6</code> , <code>turn = +0.2</code> . After checking <code>+0.8/+0.4</code> with denominator 1.0, optionally transfer to the secured robot.

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Lab Record and Exit Ticket

Split Arcade Drive

Student/team: _____ Date: _____ Robot: _____

BEFORE PLAY	AFTER TEST
☐ Robot secure ☐ DISABLED ☐ Area clear ☐ POWERING ON called ☐ Wait completed ☐ Visual clear	☐ STOP pressed ☐ DISABLED visually confirmed Safety lead initials: _____

Split-arcade mixing and normalization

DRIVE / TURN	RAW LEFT / RIGHT	DENOMINATOR	FINAL L / R + MOTION
+0.5 / 0.0			
0.0 / +0.5			
+0.6 / +0.2			
+1.0 / +1.0			
-0.5 / 0.0			
Student case: ____ / ____			

Debrief

#	QUESTION	MY RESPONSE
1	What are the two driver inputs?	
2	Why can raw mixes reach magnitude 2.0?	
3	What does the denominator guarantee?	
4	What relationship must normalization preserve?	

Exit ticket

ANSWER BEFORE LEAVING For drive = +0.6 and turn = +0.2, calculate leftPower and rightPower and state whether normalization is needed.
